



$$\bar{A} \times \bar{B} \leq \overline{A \times B}$$

3.1.9 Multisets:

Multiset is an unordered collection of elements where one element can occur more than once.

Eg: $\{m_1.a_1, m_2.a_2 \dots m_n.a_n\}$

Where m_1 is called multiplicity of set

3.1.10 Operations on multisets:

- Union: Maximum of multiplicities is considered.
- Intersection: Minimum of multiplicities is considered
- Difference: Difference of multiplicities is considered.

If the difference is -ve, it is considered 0.

- Sum: Sum of multiplicities is considered.

It is denoted as $P + Q$

3.2 Relations:

A binary relation from set A to set B is any subset of $A \times B$.

If $|A| = M$ and $|B| = n$, then $|A \times B| = mn$

$\therefore$ number of relations possible from A to B = 2^{mn}

If relation B from A to A we say it relation on A.

$\therefore$ Number of relations possible on a set A = 2^{n^2} , $|A| = n$

Domain of relation : $\{x \mid (x, y) \in R\}$

Range of relation : $\{y \mid (x, y) \in R\}$

- $(x, y) \in R$ is written as 'xRy' on 'k' relates y

3.2.1 Inverse of relations:

Inverse of a relation, $R^{-1} = \{(y, x) \mid (x, y) \in R\}$

3.2.2 Diagonal relation:

Diagonal relation on set A is

$$\Delta_A = \{(a, a) \mid \forall a \in A\}$$

3.2.3 Reflexive relations:

Relation R defined on set A is reflexive

$$\Leftrightarrow aRa \quad \forall a \in A$$

i.e., if R is a reflexive relation then $\Delta_A \leq R$

3.2.4 Important type of relations:

Type of relation	Condition	No. of relatives possible	Union	Intersection
Reflective	$aRa, \forall a \in A$	2^{n^2-n}	✓	✓
Irreflective	$\forall a \in A (a \not R a)$	2^{n^2-n}	✓	✓
Symmetric	$\forall x, y \in A (xR_y \Rightarrow yR_x)$	$2^n \cdot 2^{\frac{n^2-n}{2}}$	✓	✓
Antisymmetric	$\forall x, \forall y (xR_y \wedge yR_x \Rightarrow x = y)$	$2^n \cdot 3^{\frac{n^2-n}{2}}$	×	✓
Asymmetric	$\forall x, \forall y (xR_y \Rightarrow y \not R_x)$	$3^{\frac{n^2-n}{2}}$	×	✓
Transitive	$\forall x, \forall y (\forall z (xR_y \wedge yR_z \Rightarrow xR_z))$	—	×	✓

- Note that all the above relations are defined on a single set.
- Also to prove any relation is not of certain type we need to P.T the logical formula for that relation is false.
- Every Asymmetric relation is antisymmetric relation.

3.2.5 Composition of relations:

If R is a relation from A to B , S is a relation from B to C , the composition of R & S is given by

So R from A to C .

$$S \circ R = \{(a, c) | (a, b) \in R \text{ and } (b, c) \in S\}$$

If R is relation on A , then composition is denoted as R^2, R^3 .

Note:

If R is any relation on set A , then

$$R \circ \Delta_A = R \text{ and } \Delta_A \circ R = R$$

3.2.6 Closure:

Closure of a relation R under given property is smallest possible relation that contains R and Satisfy the property.

(i) **Reflexive Closure (R^*):**

Reflexive closure of a relation R ,

$$R^* = R \cup \Delta_A$$

(ii) **Symmetric closure:**

$$R^+ = RUR^{-1}$$

(iii) **Transitive closure:**

- Finding transitive closure has no formula, but we have a procedure as shown below.

Steps: Represent relation R with a directed graph such that whenever aRb draw a directed edge from a to b.

Steps: Now from each vertex, find all reachable vertices and for each reachable vertex b from a add the ordered pair (a, b) to the closure.

- The standard procedure for finding transitive closure is

$$R^+ = R \cup R^2 \cup R^3 \cup \dots \cup R^{n-1} \cup R^n$$

‘n’ is a positive integer such that $R^{n-1} = R^n$

- when a relation is represent as a graph,
If $(a, b) \in R^n$, we can say that there exists an n-length path from a to b.
- when asked to find more than one closure, the order to be followed is reflexive, symmetric, transitive.
Following any other order may give redundancy.
- The closure of a relation, if exists, under a property is intersection of the relations with that property containing R.
- If R is a transitive relation,
- $R^n \leq R \quad \forall n \geq 1$
- also R^n is transitive relation.

3.2.7 Partion:

A partition of set s is dividing s in disjoint subsets such that

$$A_1 \cup A_2 \cup \dots \cup A_n = s$$

$$A_i \cap A_j = \phi \quad \forall i, j \leq n$$

3.2.8 Refinement:

Partition P_2 is called refinement of partition p_1 .

if every partitioned subset of P_2 is subset to some partitioned subset of P_1 .

3.2.9 Equivalence Relation:

A relation which is reflexive

Symmetric

Transitive

is called an equivalence relation.

- Equivalence relation creates partition.
- Each set of the partition is called an equivalence class.
- Every element of same equivalence class are related to each other.
- No two element of different classes of related to each other

- Equivalence class is represented by $[a]_R$

Where a is any element of the equivalence classes

- If R is an equivalence relation, the below two sentences means the same

- aRb
- $[a]_R = [b]_R$

aRb & $[a]_R \cap [b]_R \neq \emptyset$ also mean same.

If R is an equivalence relation.

aRb is read as “ a is equivalent to b ”.

- If R_1, R_2 are two equivalence relations
 $R_1 \cup R_2$ need not to be an equivalence relation
 $R_1 \cap R_2$ is an equivalence relation.

3.2.10 Finding number of equivalence relations:

- No of equivalence relations on a set with n elements is given by

$$\text{Bell number } B_n = \sum_{k=1}^n s(n, k)$$

Where sterling 2nd kind number

$$s(m, n) = \frac{1}{n!} \sum_{i=0}^n (-1)^i n C_i (n-i)^m$$

The below is a shortcut to find the Bell number

$$B_0 \rightarrow (1)$$

$$B_1 \rightarrow (1) \quad 2$$

$$B_2 \rightarrow (2) \quad 3 \quad 5$$

$$B_3 \rightarrow (5) \quad 7 \quad 10 \quad 15$$

$$B_4 \rightarrow (15) \quad 20 \quad 27 \quad 37 \quad 52$$

Find no. of equivalence relations is nothing but find no. of way we can partition the set.

3.3 Partial Ordering Relations:

A relation R on a set A is called a partial order if R is reflexive, antisymmetric and transitive.

3.3.1 Poset:

A set ‘ A ’ together with a partial order R is called a poset

It is denoted as $[A; R]$. In a poset aRb is denoted as $a \leq b$. $a < b$ means $a \leq b$ and $a \neq b$.